Cyclic Freight Service Design with Cross-Dock Handling

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Abstract. Scheduled service network design must synchronize two flows: freight follows time-feasible connections, while reusable vehicles cover the weekly timetable with a finite fleet. This candidate isolates their interaction in an eight-leg periodic network with two tractors and one-slot cross-dock handling. The exact optimum is a direct out-and-back rotation costing 15. A generated GPT-5.4 program declares the feasible instance infeasible after duplicating destination demand. Two independent GPT-5-mini programs instead prove 11.5 with a design that needs three staggered tractors. Exact enumeration certifies both errors.

Keywords. service network design; cross-docking; periodic time-space network; fleet balance; mixed-integer linear programming

Form domain: Supply chain and logistics • Model: MILP • Exact oracle: all 2^8 service-leg designs

1. Business Brief

Business-level description. *A freight carrier publishes the same four-slot line-haul plan every week. Each operated leg moves one owned tractor; unlisted deadheads and rentals are unavailable, but tractors may wait freely at terminals. At most two tractors cover the repeating timetable. One indivisible load is ready at O in slot 0 and is due at D by slot 3. At hub H, unloading, sorting, and reloading occupy one complete slot: a load arriving in slot 1 is ready only in slot 2. Choose the weekly service legs and load path that minimize operating cost.*

The request is a compact scheduled service network design problem: vehicle conservation alone does not certify that the load can execute the same timetable.

2. Periodic Network and Instance

Let $N = \{O, H, D\} \times \{0, 1, 2, 3\}$ be periodic time-space nodes, A the listed tractor legs, and B the zero-cost one-slot waiting arcs. A return leg enters its stated node in the next weekly cycle. Each service leg has frequency at most one.

Table 1. Candidate service legs. Return legs are cargo-ineligible.

Leg	Tail	Head	Cost	Cargo
$F1$	$O0$	$H1$	2	yes
$R1$	$H1$	$O0$ next cycle	2	no
$F2$	$H1$	$D2$	2	yes
$R2$	$D2$	$H1$ next cycle	2	no
$F3$	$H2$	$D3$	10	yes
$R3$	$D3$	$H2$ next cycle	10	no
$F4$	$O0$	$D3$	7.5	yes
$R4$	$D3$	$O0$ next cycle	7.5	no

The one-slot handling transition connects the freight state $H1$ to $H2$. Thus the correct feasible cargo paths are

$$\mathcal{R} = \{r^{\text{dir}} = (F4),\ r^{\text{late}} = (F1, F3)\}.$$

The tempting pair $(F1, F2)$ is excluded: $F1$ arrives at 1, handling finishes at 2, and $F2$ departs at 1.

3. Exact MILP

Let $y_a \in \{0, 1\}$ indicate that service leg a operates, $w_b \in \mathbb{Z}_+$ count tractors on waiting arc b, and $z_r \in \{0, 1\}$ select cargo path $r \in \mathcal{R}$. Write d_e for arc duration, $H = 4$ for the weekly horizon, and $M = 2$ for the fleet. The exact formulation is

$$\min \quad \sum_{a \in A} c_a y_a, \tag{1}$$

$$\text{s.t.} \quad \sum_{r \in \mathcal{R}} z_r = 1, \tag{2}$$

$$z_r \le y_a \qquad \forall r \in \mathcal{R},\ a \in r, \tag{3}$$

$$\sum_{a \in A \cap \delta^+(n)} y_a + \sum_{b \in B \cap \delta^+(n)} w_b = \sum_{a \in A \cap \delta^-(n)} y_a + \sum_{b \in B \cap \delta^-(n)} w_b \qquad \forall n \in N, \tag{4}$$

$$\sum_{a \in A} d_a y_a + \sum_{b \in B} d_b w_b \le HM = 8, \tag{5}$$

$$y_a, z_r \in \{0, 1\}, \quad w_b \in \mathbb{Z}_+. \tag{6}$$

Equation (4) closes the periodic tractor circulation. Equation (5) is essential: total tractor-time per period cannot exceed two tractor-weeks. Path eligibility is generated only after release, deadline, and the connection rule $\mathrm{dep}(b) \geq \mathrm{arr}(a) + h_H$ through H, where $h_H = 1$.

4. Exact Outcome and Diagnostic Ablations

The direct path opens $F4, R4$ and costs 15. The legal transfer path is coverable by $F1$, one hub wait, $F3$, and $R4$, but costs 19.5. Thus the unique exact optimum is direct. Five exhaustive cases localize the two principal modeling layers.

Table 2. Correct model and four diagnostic ablations.

Model	Selected cargo/service legs	Objective
Correct handling, circulation, fleet	$F4$; $F4, R4$ (1 tractor)	**15**
Omit handling only	$F1, F2$; $F1, F2, R1, R2$ (2)	8
Omit fleet limit only	$F4$; $F4, R1, R2$ (3)	11.5
Omit all asset constraints	$F4$; $F4$	7.5
Omit handling and all assets	$F1, F2$; $F1, F2$	4

Failure certificate. A GPT-5.4 (2026-03-05, low reasoning) program returns INFEASIBLE. Its shipment balance assigns supply +1 at $O0$ but demand -1 at both $D2$ and $D3$, so total demand is two for one load. The explicit design $F4, R4$ carries the load directly, uses one tractor, and costs 15; infeasibility is therefore false.

A separate generated Gurobi model proves 11.5 with $F4, R1, R2$. Its modulo-week walk uses $F4$, three D-waits, $R2$, and $R1$: $3 + 3 + 3 + 3 = 12$ tractor-slots. Repeating its weekly departures needs $12/4 = 3$ staggered tractors, but only two exist. The underestimation is 3.5, or 23.33% of the correct optimum.

Two independent GPT-5-mini (2025-08-07, high reasoning) runs on the final canonical prompt produced executable models with the same infeasible design and objective 11.5. One collapses the winding cycle into a named-tractor circulation; the other undercounts cross-week arcs. Contemporaneous high-reasoning runs of GPT-5.4, GPT-5.5, and GPT-5.6 Luna returned the correct objective 15. The package therefore records distinct, reproducible errors without claiming universal failure.

5. Reproducibility and Sources

The repository supplies the original synthetic instance, exact Gurobi model, independently generated shipment paths, a 2^8 standard-library oracle, four ablations, the exact evaluation prompts, executable screened output, cleaned execution record, ground truth, and hashes. No upstream data or code is copied. Public package: `https://github.com/Zhengzhong-You/OR-Bench`.

Periodic time-space SND and vehicle design balance follow the conceptual setting of Gao, Jin, and Diao [1]. The one-slot loading, unloading, and transshipment semantics is motivated by Google's official Shipping Network Design documentation [2].

References

[1] A. Gao, X. Jin, and X. Diao. 2022. An iterative backbone algorithm for service network design problems. *Processes* 10(7):1373. doi:10.3390/pr10071373.

[2] Google for Developers. 2024. Shipping Network Design, Operations Research API. `https://developers.google.com/optimization/service/shipping/network_design`.